COIT11240 - Dashboard Design

Lecture Notes

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1 Signals and Times Systems

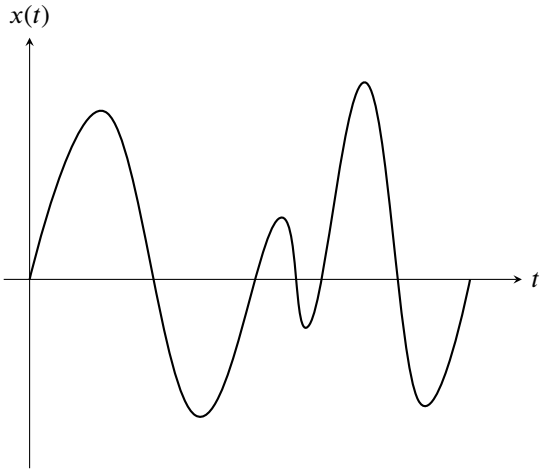
1.1 Signals:

A signal is a physical quantity that varies by something. The most common ones are time, space, and angle. Basically if it ca be measured or its effects measured, it can be though of as a signal. For example:

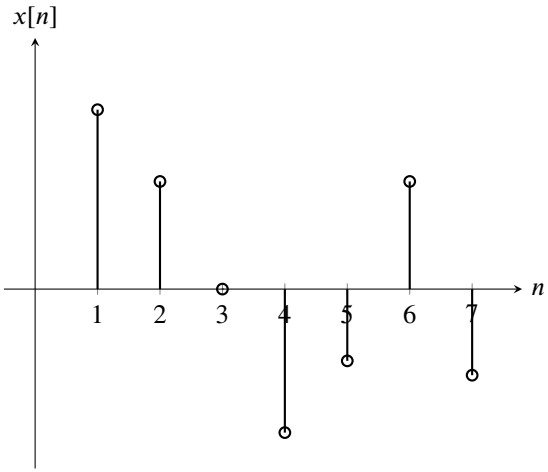
- Radio waves
- AC waveform
- Temperature
- Tide height
- Digital bit streams

1.2 Continuous vs. Discrete:

A continuous signal has a value for all time points. A waveform with considered infinite precision.



Discrete signalling on the other hand, only has values at set time points.



1.3 Sampling:

A discrete signal can be made by sampling a continuous signal of set time intervals. The sampled frequency must be twice the continuous frequency, as a minimum. The high the sampled frequency, the more accurate the output waveform.

1.4 Nyquist Theorem:

If a function $x(t)$ contains a highest frequency of BHz. For it too be completely determined sampling must be done at $\frac{1}{2B}$s.

1.5 System:

A group of regular items interacting to form a while. Such as, power amp, transformer, or laptop.

1.6 Signal Transformation:

1.6.1 Amplitued Transform

Scaling the signal

$$x(t) \rightarrow ax(t)$$

1.6.2 Amplitude Shift

Vertical shift

$$x(t) \rightarrow x(t) + b$$

1.6.3 Time Reversal

A flip across the y-axis

$$x(t) \rightarrow x(-t)$$

1.6.4 Time Scaling

Changing the frequency

$$x(t) \rightarrow x(bt)$$

- If $b > 1$: Compressing the signal
- Else $0 < b < 1$: Expanded the signal

1.6.5 Time Shifting

Horizontal movement

$$x(t) \rightarrow x(t - t_0)$$

- If $t_0 > 0$: Shifts signal right
- Else $t_0 < 0$: Shifts signal left

1.7 Even and Odd Signals:

Even signals are symmetric around the origin.

$$x(t) = x(-t)$$

Odd signals are asymmetric.

$$x(t) = -x(-t)$$

Any signal can be decomponsed into even and odd components.

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

Sub $t = -t$, into even components.

$$\begin{aligned} x_e(-t) &= \frac{1}{2}[x(-t) + x(-(-t))] \\ &= \frac{1}{2}[x(t) + x(-t)] \\ &= x_e(t) \\ &\therefore \text{Even} \end{aligned}$$

Sub $t = -t$, into odd components.

$$\begin{aligned} x_o(-t) &= \frac{1}{2}[x(-t) - x(-(-t))] \\ &= \frac{1}{2}[-x(t) + x(-t)] \\ &= -\frac{1}{2}[x(t) - x(-t)] \\ &= -x_o(t) \\ &\therefore \text{Odd} \end{aligned}$$

$$x(t) = x_e(t) + x_o(t)$$

1.8 Periodic Signals:

$$x(t) = x(t + T_0),\ T_0 > 0$$

This is where T_0 is the period.

1.8.1 Complex Form

$$\begin{aligned} x(t) &= e^{j\omega_0(t+T_0)} \\ &= e^{j\omega_0 t} e^{j\omega_0 T_0} \end{aligned}$$

1.8.2 Fundamental Period

$$T_0 = \frac{2\pi}{\omega_0}$$

1.9 General Complex Exponential:

$$C = |C|e^{j\phi}, \quad a = r + j\omega$$

$$\begin{aligned} x(t) &= |C|e^{j\phi(r+j\omega_0)t} \\ &= |C|e^{rt}e^{j\phi}e^{j\omega_0 t} \qquad = |C|e^{rt}\cos(\omega_0 t + \phi) + j|C|e^{rt}\sin(\omega_0 t + \phi) \end{aligned}$$

1.10 Step Function:

Switching operations require abrupt changes in $v(t)$ and t. At the switching point, it is undefined.

$$Ku(t) = 0,\ t < 0$$
$$Ku(t) = K,\ t > 0$$
$$Ku(t) = und\,f,\ t = 0$$

1.11 Impulse Function:

This function has infinite magnitude and zero duration. No practical application contains this, as nothing last with zero time. It is used to define derivative at discontinuity point. Discrete signals can be saved to be made up of multiple impulse functions.

$$x[n] = \sum_{k=-\inf}^{k=\inf} x[k]\delta[n-k]$$

1.12 Signal Energy and Power:

1.12.1 Continuous

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt$$

$$P = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} |x(t)|^2 dt$$

$$E_\infty = \lim_{T\to\infty}\int_{-T}^{T} |x(t)|^2 dt$$

$$P_\infty = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} |x(t)|^2 dt$$

1.12.2 Discrete

$$E = \sum_{n_1}^{n_2} |x[n]|^2$$

$$P = \frac{1}{n_2 - n_1} \sum_{n_1}^{n_2} |x[n]|^2$$

$$E_\infty = \lim_{N \to \infty} \sum_{-N}^{N} |x[n]|^2$$

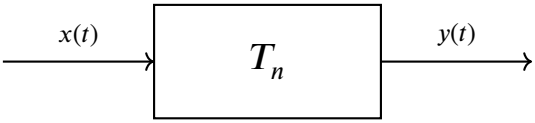
$$P_\infty = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{-N}^{N} |x[n]|^2$$

1.12.3 Fundamentals

If energy over an infinite interval is finite, $E_\infty < \infty$, then P_{av} is 0, $P_\infty = 0$. Else $P_\infty > 0 \quad \rightarrow \quad E_\infty = \infty$. A few cases can have $E_\infty = \infty$ and $P_\infty = \infty$. One such one is $x(t) = t$.

1.13 System Representation:

A system is usually represented by a black box.



$$y(t) = T_n[x(t)]$$

The T_n is not a function but a transformation happening to the function.

1.14 Causality:

A system is casual if it only depends on present and past values.

$$y(t) = 5x(t) = 6x^2(t-1) + y(t-1)$$

1.15 Invertability:

A system is said to be invertible if the operations placed on it can be revered, giving a unique answer.

1.15.1 Invertable

$$y[n] = 2x[n]$$
$$x[n] = \frac{1}{2}y[n]$$

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau$$
$$x(t) = \frac{dy(t)}{dt}$$

1.15.2 Non-Invertable

$$y(t) = x^2(t)$$
$$x(t) = \pm\sqrt{y(t)}$$

Produces 2 possible results

$$y[n] = nx[n]$$
$$x[n] = \frac{1}{n}y[n]$$

Non result if $n \neq 0$

2 Linear Time Invariant Systems

2.1 What are They?:

A linear system is one which obeys the laws of linearity and superposition:

- If x is multiplied by k making kx, $f(x)$ is also now $kf(x)$.
- When more than one factor is applied it is the sum of individual responses to each input.

2.2 Time Invarience:

Behaviour and characteristic of the system is fixed over time. An example of this is an RC circuit characteristics will not change if the experiment is conducted today, tomorrow or yesterday. Mathematically, if the input is shifted by

T, the output is also shifted by T.

$$x(t) \Rightarrow y(t)$$

$$x(t-T) \Rightarrow y(t-T)$$

2.3 Discrete Signals as Impulse Functions:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

For any given value of n, only 1 impulse in non-zero, and the rest are zero. Assume the response of an LTI system to unit impulse $\delta[n]$ is $h[n]$. Thus $\delta[n-k]$ is $h[n-k]$.

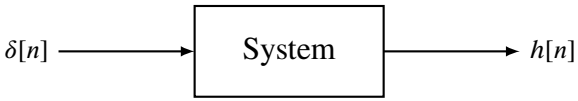
2.4 Convolution of Discrete Signals:

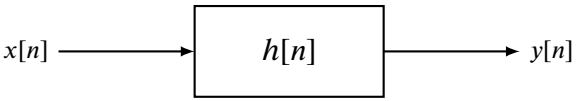
The response of that system to signal $n[n]$ is superposition of a scaled and time shifted $h[n]$.

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

This is often called the convoluetion sum.

$$\begin{aligned} y[n] &= x[n] \otimes h[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \end{aligned}$$





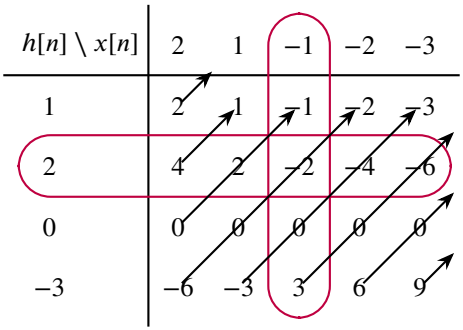
2.5 Convolution Sum-Table Method:

Let:

$$x[n] = 2, 1, -1, -2, -3$$

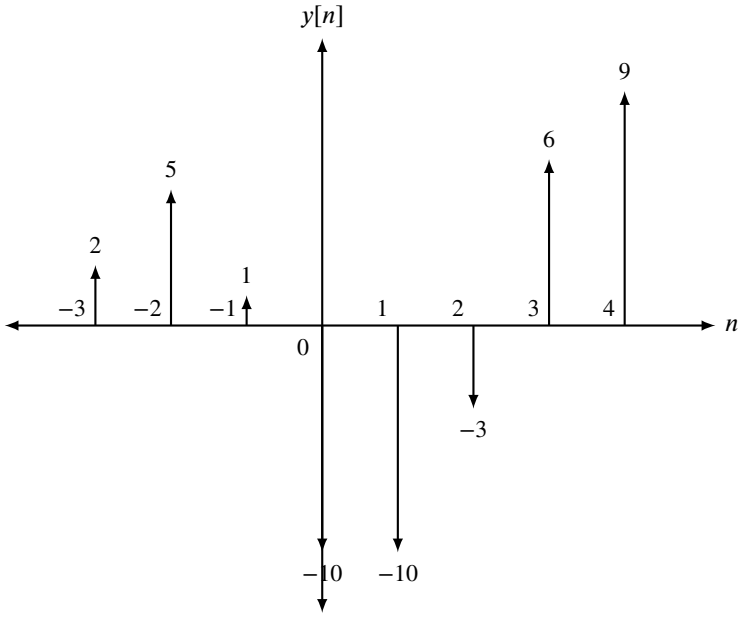
$$h[n] = 1, 2, 0, -3$$

$$x[0] = -1, \text{ and } h[0] = 2$$



$$y[n] = 2, 4, 1, -10, -10, -3, 6, 9$$

$$y[0] = -10$$



2.6 Convolution Sums Graphical Representation:

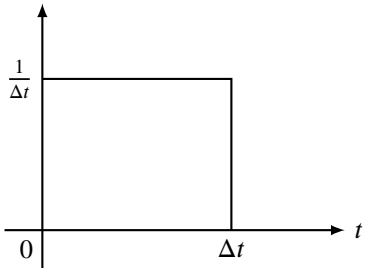
- Draw input signal as a function of k
- Draw impulse responses as function of k
- Time reverse the impulse response
- Shift impulse along t-axis
- For each n take the sum of product between input and shifted overlaps

2.7 Continuous Time Signals:

Any continuous time signal can be approximated by a thin set of strips, known as staircase approximations. The height of the strip, $\hat{x}(t)$, is the value of $x(t)$ at the start of the strip.

2.8 Pulse Function:

$$p(t) = \left\{ \begin{array}{l} \frac{1}{\Delta t},\ 0 < t < \Delta t \\ 0,\ \text{Elsewhere} \end{array} \right\}$$



$$\int_{-\infty}^{\infty} p(t)dt = t$$

2.9 Continuous Time Series Approximation:

This can be approximated as a series of pulse functions.

- Pulse located at integral multiples of pulse width (Δt)
- Pulse height is $x(\Delta t)$

Pulse of $t = \Delta t$.

$$x(k\Delta t)[p(t - k\Delta t) \cdot \Delta t] = x(\Delta t)$$

The signal can then be approximated as the sum of these signals.

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta t)[p(t - k\Delta t) \cdot \Delta t]$$

As $\Delta t \to 0$, $p(t)$ approaches an impulse function.

$$x(t) = \int_{\infty}^{\infty} x(\tau)\delta(t - \tau)d\tau$$

As $\Delta t \to 0$, $k\Delta t \to \tau$, $\Delta t \to d\tau$ and $p(t - k\Delta t) \to \delta(t - \tau)$.

2.10 Pulse Response:

If $h(t)$ is the response to the LTI system, $h_p(t)$ can be defined to be the response of a pulse system.

3 Laplace Transformation

3.1 Why Laplace?:

- Solving linear circuits frequently involves solving linear differential equations. These include transient analysis, AC analysis and transient or steady state system response.
- Finding time domain convolution
- Signals with many freqency components may be more apparent in a different domain

3.2 Laplace Transform:

$$\mathcal{L}\{f(t)\} = F(s) = \int_{0}^{\infty} f(t)e^{-st}dt$$

- Convert a time domain function to frequency domain. $t \Rightarrow s,$ where $s = \sigma + j\omega$
- Important Notes
  - If intergrand does not converge there is no Laplace transform
  - Only currently $t > 0^{-}$

3.3 Conventions:

Time domain		Frequency Domain
$f(t)$	$\Rightarrow$	$F(s)$
$g(t)$	$\Rightarrow$	$G(s)$

Functional transform is only specific functions. Operation transofmr is the description of how mathematical operations are done using Laplace.

3.4 Integration by Parts:

$$\int_0^\infty uv'dt = [uv]_0^\infty - \int_0^\infty u'vdt$$

Make sure that u simplifies when derived and v' simplifies when integrated. Laplace transform of 1:

$$\begin{aligned}
\mathcal{L}\{f(t)\} = F(s) &= \int_0^\infty f(t)e^{-st}dt \\
\mathcal{L}\{1\} &= \int_0^\infty e^{-st}dt \\
&= [-\frac{1}{s}e^{-st}]_0^\infty \\
&= [0] - [-\frac{1}{s}] \\
&= \frac{1}{s}
\end{aligned}$$

$$\begin{aligned}
f(t) &\Leftrightarrow F(s) \\
1 &\Leftrightarrow \frac{1}{s}
\end{aligned}$$

Laplace transform of t:

$$\mathcal{L}\{t\} = F(s) = \int_0^\infty te^{-st}dt$$

$$\text{Let } u = t, \quad v' = e^{-st}$$

$$u' = 1, \quad v = [-\frac{1}{s}e^{-st}]_0^\infty$$

$$\begin{aligned} F(s) &= uv - \int u'vdt \\ &= [t(-\frac{1}{s})]_0^\infty - \int_0^\infty -\frac{1}{s}e^{-st}dt \\ &= 0 + \frac{1}{s}\int_0^\infty e^{-st}dt \\ &= \frac{1}{s}[-\frac{1}{s}e^{-st}]_0^\infty \\ &= -\frac{1}{s^2}[0-1] \\ &= \frac{1}{s^2} \end{aligned}$$

$$t^n \Leftrightarrow \frac{n!}{s^{n+1}}$$

Laplace transform of $\delta(t)$:

$$\begin{aligned} \mathcal{L}\{(\delta(t)\} = F(s) &= \int_{0^-}^\infty \delta(t-0^-)e^{-st}dt \\ &= e^{-s\cdot 0} \\ &= 1 \end{aligned}$$

$$\delta(t) \Leftrightarrow 1$$

In general

$$\mathcal{L}\{\frac{d^n[\delta(t)]}{dt^n}\} = s^n$$

3.5 Operational Transforms:

3.5.1 Multiplication by a Constant

$$\begin{aligned}\mathcal{L}\{f(t)\} &= F(s) = \int_0^\infty f(t)e^{-st}dt \\ \mathcal{L}\{Kf(t)\} &= \int_0^\infty Kf(t)e^{-st}dt \\ &= K\int_0^\infty f(t)e^{-st}dt \\ &= KF(s)\end{aligned}$$

3.5.2 Addition and Subtraction

$$\begin{aligned}\mathcal{L}_1\{f_1(t)\} &= F_1(s) \\ \mathcal{L}_2\{f_2(t)\} &= F_2(s)\end{aligned}$$

$$\begin{aligned}\mathcal{L}\{f_1(t) \pm f_2(t)\} &= \int_0^\infty (f_1(t) \pm f_2(t))e^{-st}dt \\ &= \int_0^\infty f_1(t)e^{-st}dt \pm \int_0^\infty f_2(t)e^{-st}dt \\ &= F_1(s) \pm F_2(s)\end{aligned}$$

3.5.3 Differentiation

$$\mathcal{L}\{\frac{df(t)}{dt}\} = \int_{0^-}^\infty \frac{df(t)}{dt}e^{-st}dt$$

Let: $u' = \frac{df(t)}{dt}$ $\quad v = e^{-st}$

$u = f(t)$ $\quad v' = -se^{-st}$

$$\begin{aligned}\mathcal{L}\{\frac{df(t)}{dt}\} &= uv - \int uv' \\ &= [f(t)e^{-st}]_{0^-}^\infty - \int -sf(t)e^{-st}dt \\ &= \int_{0^-}^\infty f(t)se^{-st}dt - f(0^-) \\ &= sF(s) - f(0^-)\end{aligned}$$

In General:

$$\mathcal{L}\{\frac{d^n f(t)}{dt^n}\} = s^n F(s) - s^{n-1} f(0^-) - s^{n-2}\frac{df(0^-}{dt} - s^{n-3}\frac{d^2 f(0^-)}{dt^2}$$
$$- \ldots - \frac{d^{n-1} f(0^-)}{dt^{n-1}}$$

3.5.4 Integration

$$\mathcal{L}\{\int_{0^-}^{t} f(x)dx\} = \int_{0^-}^{\infty}[\int_{0^-}^{t} f(x)dx]e^{-st}dt$$
$$\text{Let: } u' = e^{-st} \quad v = \int_{0^-}^{t} f(x)dx$$
$$u = -\frac{1}{s}e^{-st} \quad v' = f(t)$$
$$\mathcal{L}\{\int_{0^-}^{t} f(x)dx\} = [-\frac{e^{-st}}{s}\int_{0^-}^{t} f(x)dx]_{0^-}^{\infty} - \int_{0^-}^{\infty} -f(t)\frac{e^{-st}}{s}dt$$
$$\lim_{t\to\infty} -\frac{e^{-st}}{s} \to 0$$
$$\lim_{t\to 0^-} \int_{0^-}^{t} f(x)dt \to 0$$
$$\therefore \mathcal{L}\{\int_{0^-}^{t}\} = \int_{0^-}^{\infty} f(t)\frac{e^{-st}}{s}dt$$
$$= \frac{1}{s}\int_{0^-}^{\infty} f(t)e^{-st}dt$$
$$= \frac{F(s)}{s}$$

3.5.5 Translation in Time Domain

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

3.5.6 Translation in Frequency Domain

$$\mathcal{L}\{e^{-at}f(t)\} = F(s+a)$$

3.5.7 Scale Change

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F(\frac{s}{a})$$

3.5.8 Frequency Differentiation

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}$$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}$$

3.6 Inverse Transform:

Once the time domain problem is solved using Laplace transformation, it needs to be converted back to time domain to understand what has happenend.

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1}s^{n-1} + \ldots + a_1 s^1 + a_0 s^0}{b_m s^m + b_{m-1}s^{m-1} + \ldots + b_1 s^1 + b_0 s^0}$$

As such, $F(s)$ is needed to be converted back into a format which contains Laplace transformations of known functions.

3.7 Partial Fraction Expansion:

If $m > n$, a proper rational function expansion. Else $m \leq n$, is improper rational functions.

$$\begin{aligned} F(s) = \frac{N(s)}{D(s)} &= \frac{s+6}{s^4 + 5s^3 + 7s^2 + 3s} \\ &= \frac{s+6}{s(s+3)(s+1)^2} \\ &= \frac{K_1}{s} + \frac{K_2}{s+3} + \frac{K_3}{s+1} + \frac{K_4}{(s+1)^2} \end{aligned}$$

$$Ke^{-at}u(t) \Leftrightarrow \frac{K}{s+a}$$

$$f(t) = K_1 u(t) + K_2 e^{-3t}u(t) + K_3 e^{-t}u(t) + K_4 te^{-t}u(t)$$